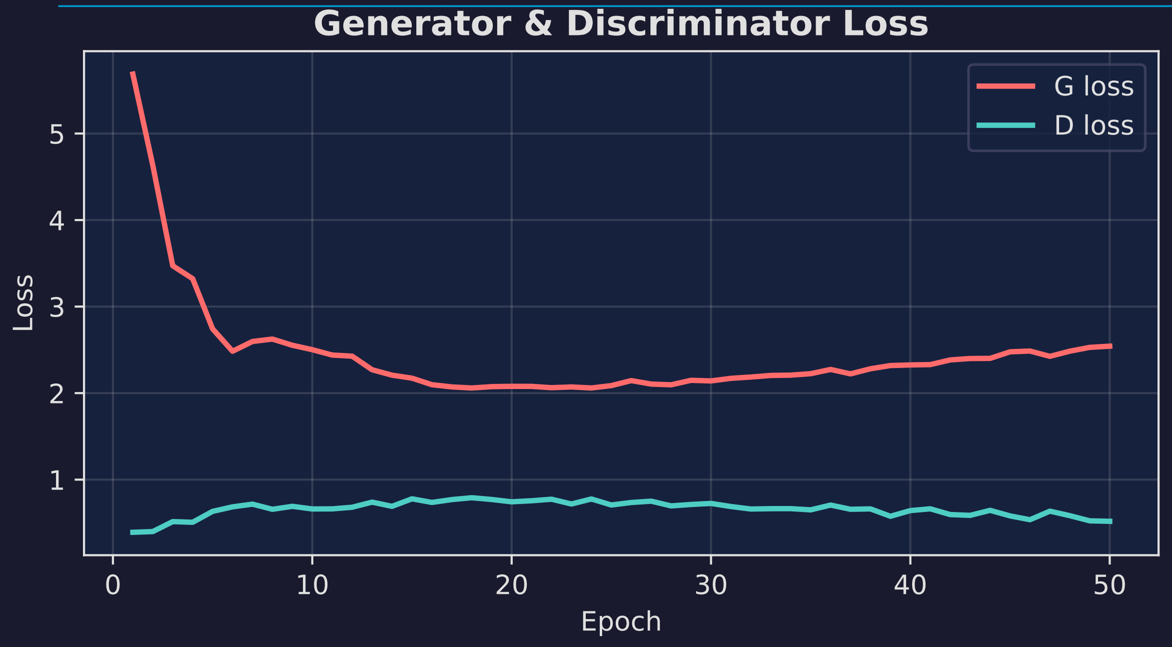
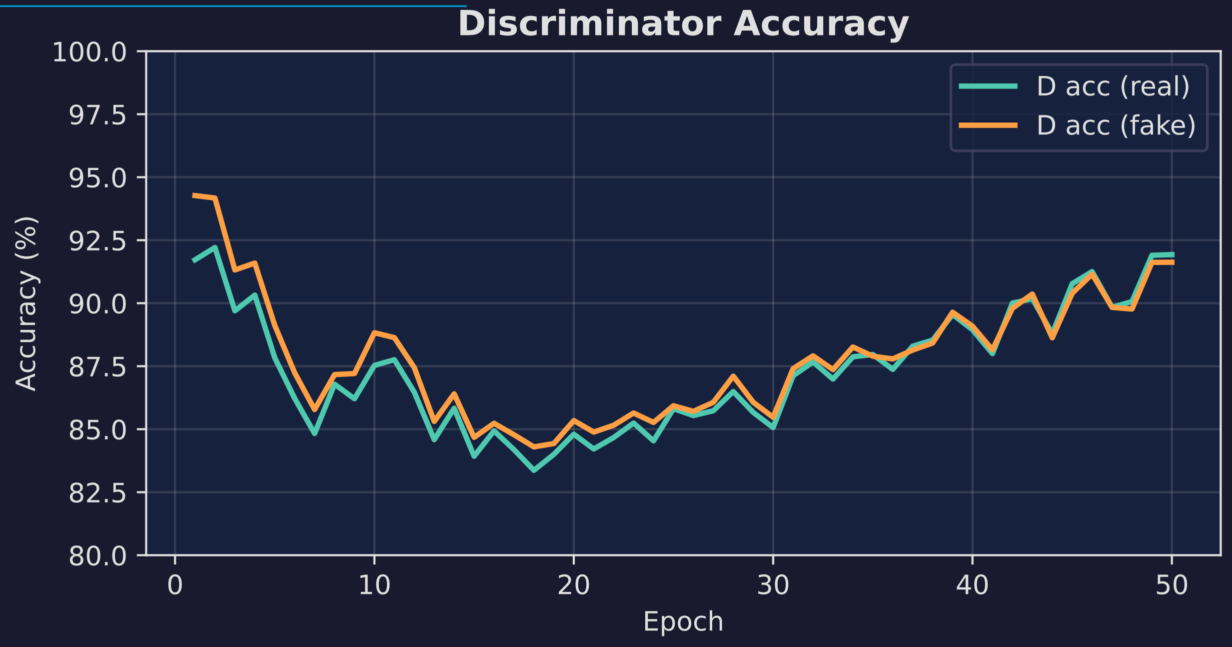
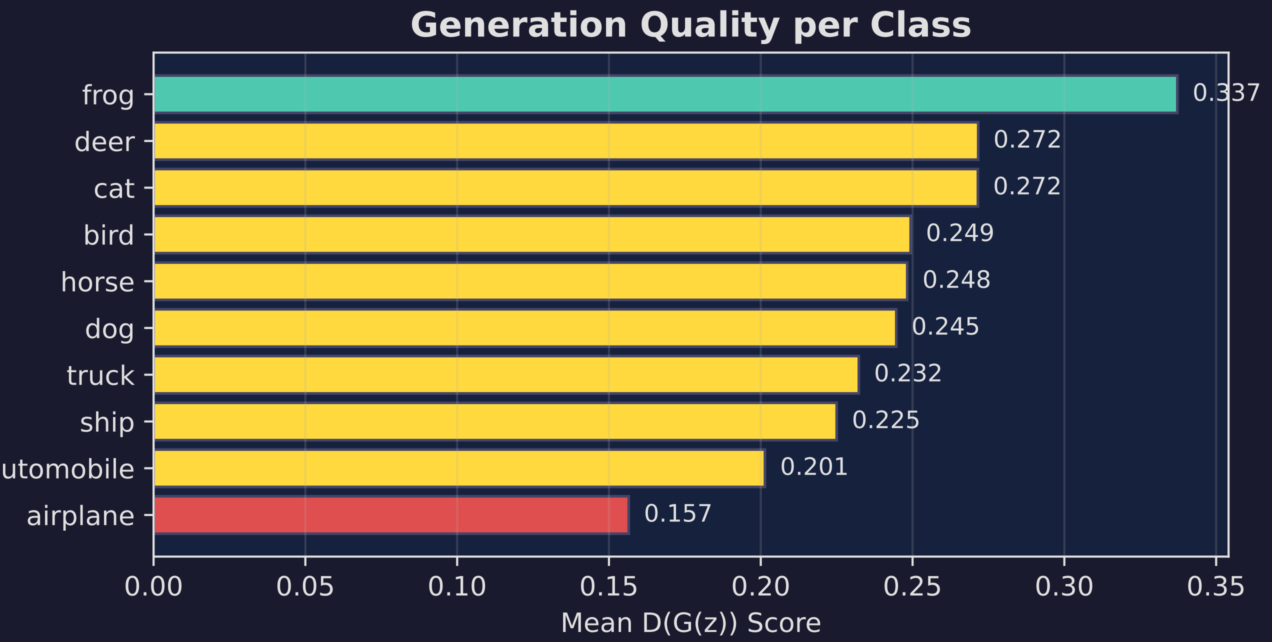
Baseline vs Improved: Configuration Comparison

Parameter	Baseline	Improved
Architecture	Vanilla DCGAN	**SNGAN ResNet**
G channels	[256, 128, 64]	**[512, 256, 128]**
D channels	[64, 128, 256]	**[128, 256, 512]**
Latent dim	100	**128**
Embed dim	50	**64**
Loss	Vanilla BCE	**Hinge loss**
D normalization	None	**Spectral Norm**
D:G ratio	1:1	**5:1 (d_steps=5)**
Optimizer	Adam ($\beta1$=0.5, $\beta2$=0.999)	**Adam ($\beta1$=0.0, $\beta2$=0.9)**
LR (G / D)	2e-4 / 2e-4	2e-4 / 2e-4
LR decay	None	**None (disabled)**
R1 penalty	None	**None (disabled)**
Weight decay G	0	**0 (disabled)**
EMA	No	**Yes (decay=0.999)**
Batch size	128	**64**
Epochs	50	**100**
Minibatch stddev	No	**Yes**

Improved Model: Training Curves & Evaluation Metrics







EVALUATION SUMMARY

Discriminator Accuracy

Real images: 62.5%

Fake images: 87.6%

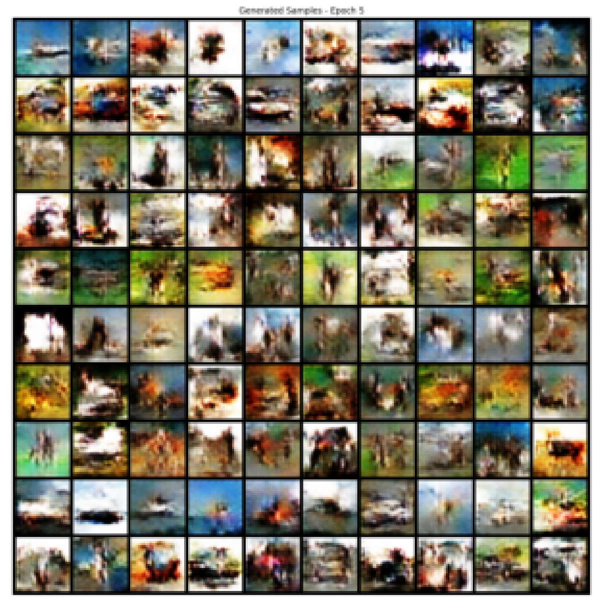
Overall: 75.0%

Training (100 epochs)

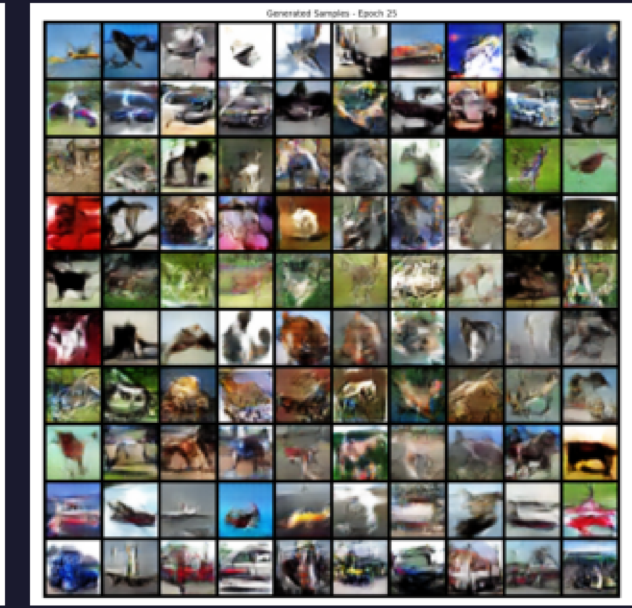
G loss: 5.69 → 2.54 (55% reduction)

Generated Samples: Training Progression

Epoch 5

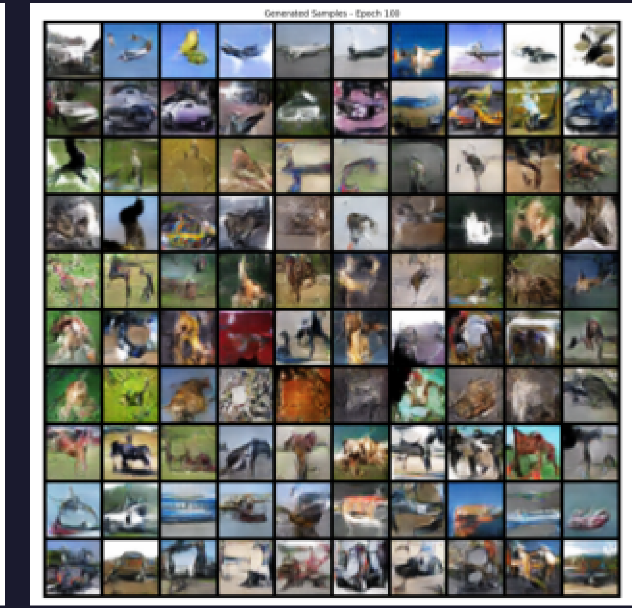


Epoch 25



Epoch 50

Epoch 100

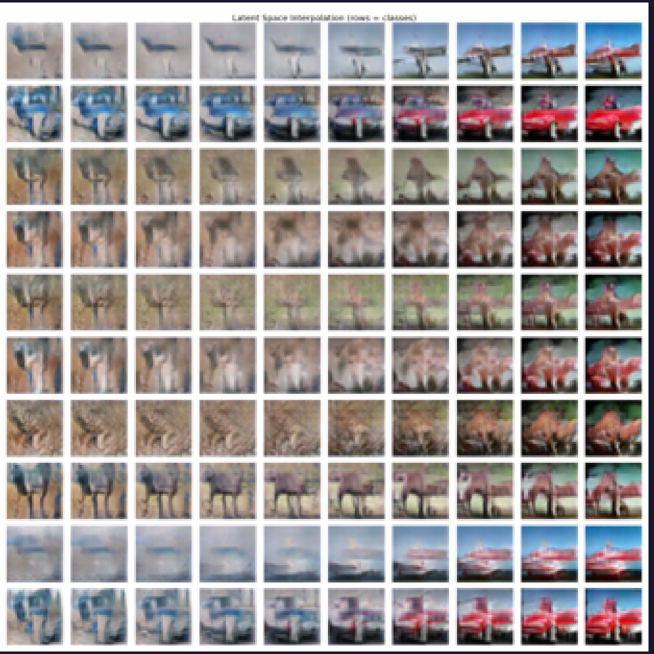


Each grid: 10 rows (CIFAR-10 classes) × 10 samples per class | Images 32×32 RGB

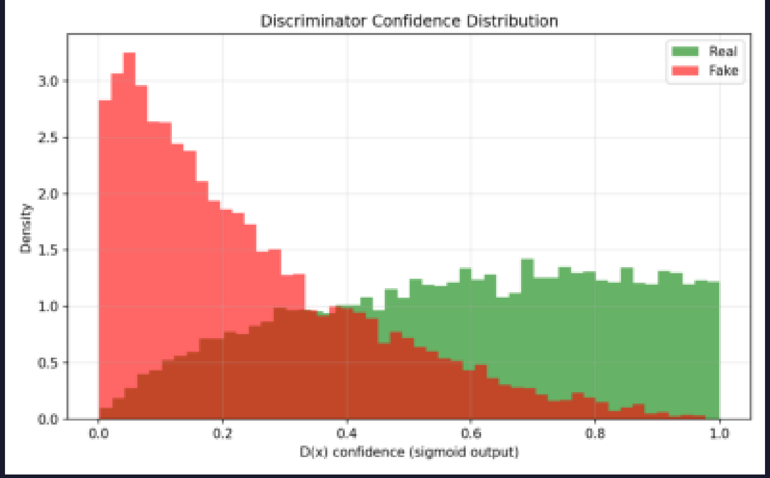
Evaluation & Comparison with State-of-the-Art

Same z, All 10 Classes

Latent Space Interpolation



D Confidence: Real vs Fake



Method	FID
Ours (improved)	**~80–120**
SNGAN (+ cBN)	21.7
SAGAN (+ attention)	18.3
BigGAN	14.7
StyleGAN2-ADA	2.4

Key Observations

- Stable training

  No mode collapse

  or divergence

- Minimal class

  differentiation

- Smooth latent

  interpolation